

Extractable surface proteins of indigenous probiotic strains confer anti-adhesion knack and protect against methicillin-resistant *Staphylococcus aureus* induced epithelial hyperpermeability in HT-29 cell line

Basavaprabhu Haranahalli Nataraj, Chette Ramesh, Rashmi Hogarehalli Mallappa *

Molecular Biology Unit, Dairy Microbiology Division, ICAR-National Dairy Research Institute, Karnal, 132001, Haryana, India

ARTICLE INFO

Keywords:

Probiotics
Surface proteins
Methicillin-resistant *Staphylococcus aureus*
Anti-adhesion
Transwell permeability

ABSTRACT

Probiotic intervention has been long believed to have beneficial effects on human health by curbing the intestinal colonization of pathogens. However, the application of live probiotics therapy may not be an ideal approach to circumvent the infections of superbug origin due to the risk of horizontal antibiotic resistance genes transfer. In this study, the anti-adhesion ability of extractable cell surface proteins from two indigenous potential probiotic strains (*Lactiplantibacillus plantarum* A5 and *Limosilactobacillus fermentum* Lf1) and two standard reference strains (*Lactobacillus acidophilus* NCFM and *Lactocaseibacillus rhamnosus* LGG) was evaluated against clinical isolates of Methicillin-Resistant *Staphylococcus aureus* (MRSA) on porcine gastric mucin and HT-29 cells. The surface proteins from the probiotic strains were extracted by treatment with 5 M lithium chloride. The surface protein quantification and SDS-PAGE profiling indicated that the yield and protein patterns were strain-specific. Surface proteins significantly hampered the mucoadhesion of MRSA isolates via protective, competitive, and displacement. Similarly, the treatment with surface proteins probiotic strains displayed anti-adhesion against MRSA isolates on HT-29 cells without affecting the viability of the cell line. Surface proteins treatment to the confluent monolayer of HT-29 cells maintained the epithelial integrity; however, MRSA isolates (10^9 cells/mL) showed considerable alteration in the epithelial integrity by exacerbating the FITC-dextran transflux. Contrarily, the co-treatment with surface proteins with MRSA isolates significantly lowered the FITC-dextran transflux across the differentiated HT-29 monolayer. Overall, the findings of this study suggest that probiotic-derived surface proteins could be the novel biotherapeutics to combat the MRSA colonization and their concomitant intestinal infections.

1. Introduction

An alarming rise in antibiotic resistance among the deadly pathogens is a major public health issue of international concern. Methicillin-Resistant *Staphylococcus aureus* (MRSA) is an important multidrug-resistant human pathogen categorized under a “global high priority list” by the World Health Organization (WHO) [1]. Recent studies on MRSA have underscored the emergence of staphylococcal resistance phenotypes towards last-line antibiotics such as daptomycin, clindamycin, and levofloxacin [2,3]. MRSA is an enduring human pathogen that colonizes on the mucous niches to evoke a variety of infections ranging from nosocomial, neonatal, and community-associated fulminant invasive infections like bacteremia, endocarditis, osteomyelitis, pneumonia, and foodborne intoxication [4]. Although anterior nares

have been traditionally considered as the primary site of MRSA colonization, there is increasing evidence on intestinal colonization of *S. aureus* which forms an important reservoir for outbreaks of resistant diseases such as antibiotic-associated diarrhea, gastroenteritis, diffuse colitis, and mild ileitis [5,6]. Besides, the direct adhesion of MRSA or their secreted toxins (enterotoxins and pore-forming toxins viz. alpha toxins and alpha-hemolysin) could disrupt the gut barrier intricate to trigger bacteremia and sepsis [7–9]. Intestinal colonization of MRSA has been observed to cause postoperative infection by translocation from gut to wound site and other visceral organs [10,11]. Moreover, there have been reports portraying intestinal colonization of MRSA due to the failure of previous mainstay antibiotics in pathogen clearance [12,13]. The colonization of such resistant bugs in the human gut may create a conducive environment for the transfer of potential antibiotic-resistant

* Corresponding author.

E-mail address: Rashmi.M@icar.gov.in (R.H. Mallappa).

<https://doi.org/10.1016/j.micpath.2021.104974>

Received 14 January 2021; Received in revised form 29 April 2021; Accepted 30 April 2021

Available online 18 May 2021

0882-4010/© 2021 Elsevier Ltd. All rights reserved.

genes (ARGs) from MRSA to commensal bugs and thereby transforming the gut as an epicenter of antibiotic resistance. Therefore, it is crucial to reduce the use of antibiotics in infection control and to unravel the antagonistic abilities of novel biotherapeutic agents that originate from a safe source.

Accordingly, modern scientists have proposed several alternatives to antibiotics or adjunct avenues to avert the aggravating antibiotic resistance crisis. These include probiotics, prebiotics, synbiotics, bacteriocins, antimicrobial peptides, bacteriophages, endolysins, antibodies, and fecal microbiota transplantation (FMT) [14]. The efficacy of such alternatives indeed relies on the mode and site of their application. Nevertheless, few limitations of probiotics and phages have obstructed their full potential applications as alternatives to conventional antibiotics. The use of phages has not been widely accepted due to the narrow spectrum activity, the possibility of phage-mediated transfer of genetic material to the bacterial host, and regulatory issues [15]. Similarly, the use of probiotics may increase the risk of acquiring the ARGs from the superbugs through horizontal gene transfer (HGT) and the resultant transconjugants (probiotics possessing ARGs) may act as the potential vector for transferring the antibiotic resistance to the commensal microbes. Hence, exploring paraprobiotics (non-viable probiotics or their cell surface associated components) in studying the antagonistic attributes against multidrug-resistant (MDR) superbugs possibly overcomes the limitations posed by probiotics.

Paraprobiotics is relatively a novel and emerging concept of functional food that constitutes the study on functional benefits of the probiotic bacterial-derived cell wall components [16,17]. Paraprobiotics include the inactivated/dead/non-viable microbial cells of probiotics (intact or ruptured containing probiotic cell components upon lysis such as surface proteins, teichoic acids, peptidoglycan-derived muropeptides, cell surface proteins, exopolysaccharides, and surface protruding molecules such as pili, fimbriae, and flagella [17]. Bacterial surface proteins are the microbes associated-molecular patterns (MAMPs) that are attached non-covalently to the cell wall to exhibit several physiological and cellular functions of probiotics. Surface proteins assist probiotic bacteria in excreting several probiotic phenotypes such as auto-aggregation, co-aggregation, stress tolerance, bacterial adhesion to gut enterocytes to exclude pathogens, and immunomodulation [18]. It is also worth mentioning that surface proteins of probiotic *Lactobacillus acidophilus* in combination with nisin have also shown anti-staphylococcal activity by damaging the cell membrane coherence and subsequent nucleic acid leakage [19].

Adhesion to epithelial cells is crucial for any potential probiotics to demonstrate various health-promoting attributes, including pathogen elimination. Successful probiotic bacterial adhesion to intestinal epithelial cells creates colonization resistance against invading pathogens by blocking the adhesion sites. The probiotic bacterial adhesion to intestinal epithelium is assisted by its cell surface appendages including cell surface proteins. This statement has been supported by various research findings where the removal of cell surface proteins has witnessed a tremendous reduction in the probiotic bacterial adhesion to intestinal epithelial cells [20,21]. These findings were further validated by molecular biology experiments where deletion of genes coding for surface proteins had significantly lowered the epithelial (Caco-2/HCT-111/HT-29 cells) adhesion of knockout strain compared to wild strain [22–24]. These findings highlight that surface proteins are the key players in probiotic bacterial adhesion to gut enterocytes.

Owing to these reports, we hypothesized that surface proteins might be the key players in an effective probiotic bacterial adhesion to gut enterocytes, and therefore their intervention via protective, competitive, and displacement modes could lower the epithelial adhesion of MRSA. In this study, the surface proteins of two indigenous probiotics strains and two commercial probiotic strains have been used to investigate their anti-adhesion ability against MRSA on extracellular matrix (gastric mucin) and HT-29 cells. Additionally, the impact of surface proteins intervention on epithelial permeability of HT-29 cells was studied under

the MRSA-induced epithelial hyperpermeability model.

2. Materials and methods

2.1. Bacterial cultures and growth conditions

Two indigenous probiotic strains of human gut origin viz. *Lactiplantibacillus plantarum* A5 and *Limosilactobacillus fermentum* Lf1 along with two standard reference strains viz. *Lactiseibacillus rhamnosus* GG and *Lactobacillus acidophilus* NCFM were collected from the Molecular Biology Unit, Dairy Microbiology Unit, ICAR-National Dairy Research Institute, Karnal. The probiotic strains were cultured and propagated aerobically in deMan, Rogosa, and Sharpe (MRS) broth (HiMedia, India) at 37 °C/12–18 h.

On the other hand, two genotypically identified multi-drug resistant clinical MRSA isolates (12/206 and 5/255) were gifted from the All India Institute of Medical Sciences (AIIMS), New Delhi, India. These isolates exhibited preliminary features of intestinal adhesion by exhibiting superior cell surface properties (cell surface hydrophobicity and auto-aggregation) and *in vitro* mucin adhesion [25]. MRSA isolates were maintained and propagated aerobically in BHI broth (HiMedia, India) at 37 °C/14–18 h. Besides, the glycerol stocks (20%) of the isolates were stored at –80 °C. For the experimental purpose, the overnight grown active MRSA cells were harvested by centrifugation (1300×g for 10 min) and washed twice with 0.01 M phosphate-buffered saline (PBS; pH = 7.4) (Sigma, USA) and suspended in PBS to adjust the cell density to 10⁹ CFU/mL (confirmed by pour plate method).

2.2. Epithelial cell culture

The human colon adenocarcinoma HT-29 (ATCC HTB-38) cell line with unique features of gut enterocytes was previously procured from the National Center of Cell Sciences (NCCS) (Pune, India). The cells were propagated in Dulbecco's Modified Eagle's Minimal Essential Medium (DMEM-High glucose) (Sigma, USA) supplemented with 10% Fetal Bovine Serum (FBS), 2% sodium bicarbonate (Sigma, USA), 100 U/mL penicillin, and 100 µg/mL streptomycin (Sigma, China) in 25 cm² (T-25) tissue culture flask (Thermo Fisher Scientific, China). The cells were routinely maintained at 37 °C with 5% carbon-di-oxide (CO₂) atmosphere in a humidified incubator (NUAIRE, USA).

2.3. Extraction and quantification of surface proteins

The total surface proteins of probiotic strains under the study were extracted as previously described by Turner et al. [26] with slight modifications. Briefly, the probiotic strains were cultured in 1L MRS broth (37 °C for 24 h) and the cells were harvested by centrifugation (9000×g for 20 min) and washed twice with PBS (pH = 7.4). The cell pellet was suspended in 50 mL of 5 M lithium chloride (LiCl) (Sigma, USA) and incubated at refrigerated conditions for 2 h with intermittent vigorous agitation. Further, the surface proteins were recovered by centrifuging at 3200×g for 15 min. The supernatant was collected and dialyzed by 10–12 kDa molecular weight cutoff (MWCO) membrane (Sigma, USA) against sterile distilled water containing 100 mM phenylmethyl sulphonyl fluoride (PMSF) (HiMedia, India) dissolved in 1 mL isopropanol. The dialysis was carried out for two successive days at refrigerated conditions with the constant replacement of distilled water every 24 h. The obtained protein suspension was lyophilized and stored at –20 °C until further usage. On the other side, Lowry's method of protein estimation was used to quantify the surface proteins using bovine serum albumin (BSA) as standard [27].

2.4. SDS-PAGE profiling of surface proteins

Protein profiling was carried out by Laemmli SDS-PAGE (Sodium Dodecyl Sulfate Poly-Acrylamide Gel Electrophoresis) using 4% (w/v)

stacking gel and 12% (w/v) resolving gel [28]. The separated protein bands were stained using Coomassie Brilliant Blue (CBB) dye. The protein bands were visualized by scanning the resolving gel in a 2D gel documentation system (Epson Scanner; Proteomics images and spot picker, M/S GE Health care bioscience Pvt Ltd., New Delhi).

2.5. Cytotoxic effect of surface proteins and MRSA isolates on the viability of HT-29 cells

The cytotoxic effect of surface proteins and MRSA isolates on HT-29 cells was measured by trypan blue exclusion assay [29]. HT-29 cells were cultured in 12 well tissue culture plates until cells reach 90% confluency. After the monolayer formation, the cells were incubated overnight with antibiotic-free DMEM and finally rinsed thrice with PBS (pH = 7.4). Thereafter, exactly 1 mL of filter-sterilized surface proteins dissolved in DMEM (100 µg/mL) and MRSA isolates (1×10^9 CFU/mL) (suspended in the antibiotic-free-DMEM) were added to HT-29 monolayer and incubated independently at 37 °C for 24 h in 5% CO₂ incubator. At the end of incubation, the spent medium was aspirated and rinsed thrice with PBS (pH 7.4). To quantify the viability of HT-29 cells, the adhered HT-29 cells were detached by 1X (0.25%) trypsin digestion (250 µL/well) (HyClone, USA) for about 5 min at 37 °C and further cells were treated with 250 µL fresh DMEM medium to nullify the deleterious action of trypsin. An aliquot of 50 µL trypsinized suspension was mixed with an equal volume of 0.5% (v/v) trypan blue and counted using a hemocytometer (HBG, Germany). The colorless and blue colored cells were considered as viable and non-viable, respectively. The results were expressed as percentage viability using the following equation:

$$\% \text{ Viability} = \left[\frac{\text{Total number of cells per field} - \text{Number of dead cells per field}}{\text{Total number of cells per field}} \times 100 \right]$$

2.6. Surface proteins interference assays

The surface protein interferences to inhibit the MRSA adhesion was carried out according to the method of Chatterjee et al. [31] with slight modifications as described below.

2.6.1. Culturing HT-29 cells on 12-well tissue culture plates

HT-29 cells were seeded ($\sim 5 \times 10^4$ cells/well) in 12 well tissue culture plates and allowed to form a confluent monolayer (80–90% confluency) by incubating at 37 °C for 2–3 days in 5% CO₂ incubator. One day before the assay, the spent medium was replaced with antibiotic-free DMEM. Before the start of the experiment, the spent medium was removed and the cells were washed twice with PBS (pH = 7.4).

2.6.2. Preparation of mucin coated plates

One per cent (w/v) porcine gastric mucin-type III (Sigma, USA) solution was prepared by suspending the type III mucin in PBS (pH = 7.4) at 4 °C for 2 h with intermittent mixing. After incubation, the 500 µL of filter-sterilized mucin solution was coated or transferred to the 24 well tissue culture plate (Thermo Fisher Scientific, China) and incubated overnight at 4 °C. Thereafter, the plates were rinsed twice with 500 µL of PBS to remove the unbound mucin. The unevenness on the mucin surface was blocked by incubating with 500 µL of 2% BSA (Sigma, USA) at 4 °C for 4 h. The unbound fractions of BSA were removed by washing twice with 500 µL of PBS.

2.6.3. Exclusion or protective assay

The HT-29 monolayer and mucin coated wells were incubated with 1 mL filter-sterilized surface proteins suspension dissolved in PBS (100 µg/mL) at 37 °C for 2 h. After the specified incubation, the unbound fractions were removed by washing twice with PBS (pH 7.4) and

immediately challenged with 1 mL of MRSA cell suspension (10^9 CFU/mL) for 2 h at 37 °C. The unadhered cells were discarded by washing twice with PBS, and the adhered MRSA cells were recovered by lysing the cells with 1 mL triton-X-100 (0.5% v/v). The bound MRSA cells were quantified by serial dilution and subsequent plating using selective agar (BHI agar supplemented with 5 µg/mL of ampicillin) and the results were expressed as log₁₀ CFU/mL.

2.6.4. Competitive assay

The confluent monolayer of HT-29 cells and mucin wells were co-incubated with a mixture of 0.5 mL of surface proteins suspension (100 µg/mL) and 0.5 mL of MRSA cell suspension (both at 10^9 CFU/mL) for 37 °C for 4 h in 5% CO₂ incubator. The unadhered MRSA and surface proteins fractions were removed by washing twice with 1 mL of PBS (pH 7.4). Finally, the adhered MRSA cells were enumerated as described under section 2.6.3.

2.6.5. Displacement assay

In the displacement assay, 1 mL of MRSA cell suspension (10^9 CFU/mL) was pre-incubated with confluent HT-29 cells and mucin coated wells at 37 °C for 2 h. At the end of incubation, the unadhered MRSA cells were discarded by washing twice with 1 mL of PBS (pH 7.4) each time. Subsequently, the monolayer was challenged with 1 mL of surface proteins (100 µg/mL) and incubated for 2 h in 5% CO₂ incubator. Finally, the adhered MRSA to HT-29 cells were counted as explained under section 2.6.3.

2.7. Impact of probiotic surface proteins and MRSA isolates on epithelial permeability

The effect of surface proteins and MRSA isolates on intestinal epithelial barrier integrity was investigated by transwell permeability assay by measuring the Fluorescein isothiocyanate (FITC)-dextran flux across the differentiated HT-29 monolayer [31]. At the outset, HT-29 cells were seeded ($\sim 10^5$ cells/well) on the apical chamber of transwell membrane insert (0.4 µm pore size, 12 mm insert diameter) of 12 well tissue culture polystyrene plates (Corning, Sigma, USA). The apical and basal chambers were supplemented with 1 mL and 1.5 mL of DMEM, respectively. The cells were allowed to proliferate until they reach the polarized stage by incubating at 37 °C for 3–4 days in 5% CO₂ incubator.

2.7.1. Assessment of monolayer integrity of HT-29 cells

The formation of an intact monolayer of HT-29 cells was assessed before the commencement of the experiment by the dye diffusion test [32,33]. After the cells reach complete confluency on the apical membrane (examined through an inverted microscope), the spent medium was removed and rinsed twice with 0.5 mL of PBS (pH 7.4). The upper chamber of the transwell was filled with 0.5 mL of 100 µg/mL FITC-dextran (70 kDa) (Sigma, USA) whose fluorescent absorbance at 492 nm (excitation) and 520 nm (emission) was previously measured by Synergy H1 Multimode Reader. Exactly 1 mL PBS was added in the basal chamber and plates were incubated at 37 °C for 1 h in 5% CO₂ incubator at dark condition. After the specified incubation, the fluorescent absorbance of the basolateral sample was recorded as described above and culture inserts with less than 0.5% flux were selected for the next experiments.

2.7.2. Effect of surface proteins and MRSA isolates on FITC- dextran permeability in HT-29 cell line

The transwell inserts with less than 0.5% flux were incubated with 0.5 mL of test pathogens (MRSA: 10^9 CFU/mL) and probiotic-derived surface protein suspended in antibiotic-free DMEM at 37 °C for 24 h (5% CO₂, 90% humidity). On the other side, 0.5 mL of a combined mixture of surface proteins (100 µg/mL) and MRSA (10^9 CFU/mL) were co-incubated for 24 h on the HT-29 monolayer with the same incubation conditions. The basal chamber was added with 1.5 mL of DMEM

(antibiotic-free) for the continuous supply of nutrients to HT-29 cells. After aforesaid incubation, the spent medium was removed from apical and basal chambers and the monolayer was rinsed gently with 1 mL PBS. Further, the basal chamber was filled with 1 mL PBS and the apical chamber was filled with 0.5 mL (100 µg/mL) of FITC-dextran and incubated at 37 °C for 6 h. Finally, the fluorescent absorbance was measured as previously described and the per cent diffusion of FITC dextran across the HT-29 monolayer was calculated using the below equation:

$$\% \text{ Transflux} = (\text{Absorbance at basal chamber} / \text{Absorbance at apical chamber}) \times 100$$

2.8. Statistical analysis

Data were subjected to one-way ANOVA (non-parametric) following the Tukey test using GraphPad Prism software (version 5.01, USA). Differences were considered statistically significant when $p \leq 0.05$.

3. Results and discussion

Prevalence of gastrointestinal colonization of multi-drug resistant MRSA and their subsequent infections are the major therapeutic challenge as such infections are generally resistant to antibiotics. Gut microbial manipulation by the intervention of probiotics, prebiotics, and/or bacteriophages has been proposed as an adjunct therapy to combat such infections [34]. However, the concept of paraprobiotics has been recently picturized to overcome the limitations (horizontal gene transfer from pathogens to probiotics) posed by live probiotics for direct biotherapeutic applications against antibiotic-resistant superbugs. Herein, we tested probiotic bacterial-derived extractable surface proteins as alternative armamentarium to decolonize MRSA from the intestinal milieu.

3.1. Extraction and quantification of surface proteins

Surface proteins are the molecular patterns attached non-covalently to the bacterial cell wall and are involved in several host-microbial interactions [18]. They are not only involved in providing bacterial protection against gastrointestinal stress but also aids in the effective adhesion of probiotic bacteria to the intestinal epithelial cells. Hence, we were keen to extract those surface proteins from the probiotic cells and use them independently to study their anti-adhesion ability against MRSA. Treating probiotic cells with chaotropic agents, especially LiCl, shaves the non-covalently bound surface proteins [35]. In the present study, the surface proteins were extracted from 24 h grown probiotic cells by treatment with 5 M LiCl. The yield of surface proteins from indigenous probiotics and standard reference strains varied to a greater extent. Surface proteins derived from *L. fermentum* Lf1 and *L. plantarum* A5 had a protein concentration of 517.5 and 446.8 µg/mL, respectively. However, the reference strains showed a higher protein yield of 1308.5 (*L. acidophilus* NCFM) and 1033.5 µg/mL (*L. rhamnosus* GG). In agreement with our findings, Zhang et al. [36] have also reported the variations in the yield of surface proteins among the various probiotic *Lactobacillus* strains cultured with similar growth conditions. The yield of surface proteins may also depend on the chaotropic agent used to shave the proteins. In contrast to our results, Sánchez et al. [37] have quantified the surface proteins of *L. rhamnosus* GG cultured in MRS broth and found 591.7 ± 49.8 µg/mL. This observed variation in the yield of surface proteins from LGG may be attributed to the growth phase of cells used for protein extraction. The researchers harvested the 16 h grown culture for surface proteins extraction, whereas the present study exploited 24 h old cultures. Zhang et al. [36] have also demonstrated the change in the temporal expression of surface proteins with the change in

incubation conditions (temperature and aerobic/anaerobic conditions), wherein the maximum yield of surface proteins was observed with the cells cultured aerobically at 37 °C.

3.2. SDS-PAGE profiling of surface proteins

Surface proteins of lactic acid bacteria constitute several proteins such as S-layer proteins, mucin binding proteins, fibronectin-binding proteins, moonlight proteins, and so on. Indeed, they are considered to play an important role in displaying probiotic phenotypes and modulating several physiological and signal transduction functions. As such, extracted and purified surface proteins of various molecular weights have reported retrieving some of such properties viz. pathogen exclusion, immunomodulation, and gut barrier function properties [18, 37,38]. Hence, to confirm the presence of various surface proteins, the protein samples were run on SDS-PAGE. Fig. 1 represents the heterogeneity in the surface proteins expression pattern of various probiotic strains under the investigation. The surface proteins of *L. fermentum* Lf1 revealed several faint and intensified bands of size ~100, 70, 45, 37, and 30 kDa. Whereas, *L. plantarum* A5 showed dominant bands of size ~45, 50, and 37 kDa besides several light bands. On the contrary, reference strain *L. acidophilus* NCFM displayed intensified bands at 75, 45, 37, and 30 kDa besides several faint bands. Similarly, *L. rhamnosus* GG showed the prominent bands at 37, 30, and 50 kDa. These bands can be roughly correlated to the various cells surface proteins viz. mucin binding proteins, fibronectin binding proteins, collagen binding proteins, S-layer proteins, moonlight proteins, etc. Our findings are in line with the previous studies, where investigators have correlated the band at 72 kDa to collagen-binding protein in *L. plantarum* Lp91 [39], 37 kDa to glyceraldehyde-3-phosphate dehydrogenase (GAPDH) [40], 45 kDa to S-layer protein in *L. acidophilus* NCFM [41], 64 kDa to fibronectin-binding protein in *L. acidophilus* NCFM [42]. Likewise, all the strains under the study have displayed a band at 30–32 kDa with varied intensity, and thereby confirmed the expression of mucin binding protein alike the findings of Rojas et al. [38] and Macías-Rodríguez et al. [43]. It is imperative to note that several of these proteins have been reported in single or multiple cellular functions (moonlight proteins) [44]. For example, mucin binding proteins help in eliciting probiotic bacterial adhesion to dynamic mucus matrix in the colon. Fibronectin and collagen-binding proteins facilitate bacterial adhesion respectively to fibronectin and collagen, which are considered important extracellular matrix (ECM) components of eukaryotic tissues. Moonlighting proteins are the membrane-spanning proteins with more than one function besides their involvement in bacterial adhesion to epithelial cells or ECM. These include GAPDH, EnoA1, elongation factor G, pyruvate kinase pyruvate dehydrogenase (PDHB), GroEL heat shock proteins, etc. [45]. Though the present study could not able to identify and distinguish all the surface anchored protein molecules through the SDS-PAGE technique, further studies employing advanced proteomic approaches could be able to address the limitations of this study.

3.3. Cytotoxic effects of surface proteins and MRSA isolates on HT-29 cells

Safety assessment of biological molecules has given a lot of deliberations since their interventions may affect the viability and normal physiology of the eukaryotic cells by altering the cell membrane coherence, inhibition of protein synthesis, inhibition of enzymes, etc. [46]. Therefore, determining *in vitro* cytotoxicity of any biomolecule of therapeutic importance is an essential step before animal or human applications. Moreover, the cytotoxicity assessment is the only available method to check the dosage compatibility of test compounds with the cell line used in further experiments. In this study, the surface proteins of probiotic strains (100 µg/mL) had no significant change in the viability of HT-29 cells as compared to control and therefore encouraged their further applications on gut enterocytes with the same dosage. Similar to

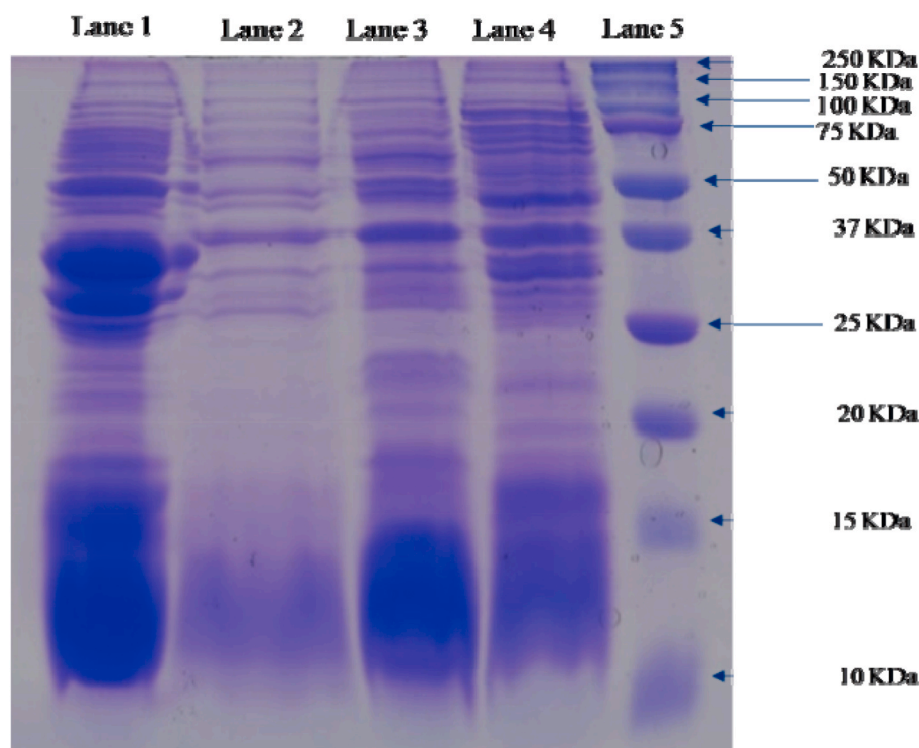


Fig. 1. SDS-PAGE profile of surface proteins extracted from probiotic strains under the study. Lane 1: *L. rhamnosus* GG, Lane 2: *L. plantarum* A5, Lane 3: *L. acidophilus* NCFM, Lane 4: *L. fermentum* Lf1, Lane 5: Protein marker.

our findings, Zhao et al. [47] have documented the non-toxic effect of surface proteins of *L. casei* zhang, *L. rhamnosus*, *L. gasseri*, and *L. acidophilus* NCFM (100 µg/mL) on HT-29 cells. On the other hand, Nozari et al. [48] reported 42% loss of viability of Caco-2 cells upon treatment with cell wall protein fractions from *Lactocaseibacillus paracasei* (100 µg/mL) for 72 h. This indicates that the toxicity of surface proteins may vary with the different concentrations, duration of exposure, and strains. MRSA treatment (10^9 cells/mL) significantly decreased the viability (~10–15%) of HT-29 cells as presented in Fig. 2. In contradiction, Jayashree et al. [4] have reported an approximate 70–75% decrease in the viability of Caco-2 cells upon exposure to MRSA (10^6 cells/mL). These findings explain the heterogeneity in virulence

patterns or aggressiveness of the different MRSA isolates.

3.4. Mucoantiadhesive action of surface proteins

Similar to the standard reference strains (NCFM and LGG), the surface proteins of indigenous probiotics viz. Lf1 and A5 significantly ($p \leq 0.01$) reduced the adhesion of MRSA 12/206 to 6.50 ± 0.21 and 6.50 ± 0.40 log₁₀ CFU/mL, respectively as shown in Fig. 3A. On the other hand, A5 and NCFM considerably lowered the adhesion of MRSA 5/255 to 6.10 ± 0.10 and 6.06 ± 0.13 log₁₀ CFU/mL respectively. Likewise, the surface proteins of Lf1 and LGG significantly ($p \leq 0.05$) blocked the mucoadhesion of MRSA 5/255 to 6.30 ± 0.10 and 6.50 ± 0.20 log₁₀ CFU/mL, respectively (Fig. 3B).

The surface proteins have competitively blocked the mucoadhesion of MRSA 12/206 (8.40 ± 0.10 log₁₀ CFU/mL) by declining the adhesion to 6.40 ± 0.40 (Lf1), 6.60 ± 0.10 (A5), 6.45 ± 0.45 (NCFM) and 6.90 ± 0.10 (LGG) log₁₀ CFU/mL (Fig. 3A). Similarly against MRSA 5/255 (8.50 ± 0.20 log₁₀ CFU/mL), the surface proteins of indigenous probiotics have decreased the mucoadhesion to log₁₀ CFU/mL values of 6.15 ± 0.15 (Lf1) and 6.30 ± 0.10 (A5) as shown in Fig. 3B.

Fascinatingly, the surface proteins of all the probiotic strains have significantly ($p \leq 0.05$) displaced the adhered MRSA 12/206 (8.25 ± 0.05 log₁₀ CFU/mL) by exhibiting the lowered adhesion to mucin 7.30 ± 0.10 (Lf1), 6.20 ± 0.20 (A5), 6.35 ± 0.05 (NCFM), and 7.20 ± 0.00 (LGG) log₁₀ CFU/mL, respectively. Amongst all, A5 and NCFM showed the highest potential to displace the mucin adhered MRSA 12/206 (Fig. 3A). Against MRSA 5/255 (8.30 ± 0.10 log₁₀ CFU/mL), the surface proteins of Lf1, A5, NCFM, and LGG have significantly mitigated adhesion of MRSA 5/255 to 7.25 ± 0.05 , 7.20 ± 0.10 , 7.00 ± 0.10 , and 7.10 ± 0.20 log₁₀ CFU/mL, respectively, as presented in Fig. 3B.

Taken together, the surface proteins of indigenous probiotic strains viz. A5 and Lf1 were found equally potent to reference strains in decolonizing MRSA isolates. However, the effective mucoadhesive action of surface proteins relies on the interaction between acidic charged surface proteins (S-layer proteins and mucus binding proteins) and the

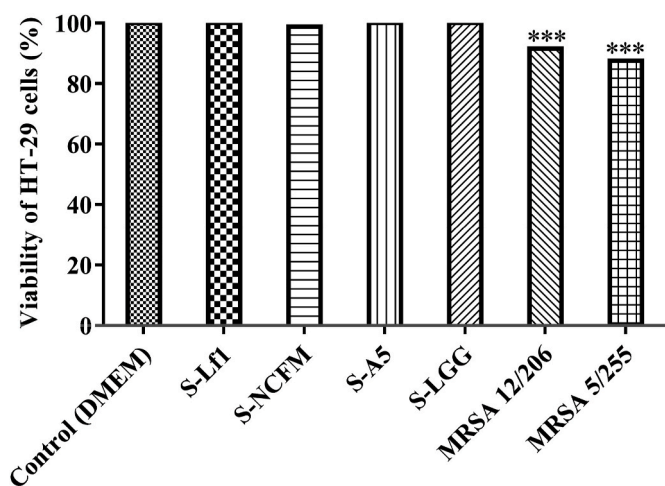


Fig. 2. Effect of probiotic surface proteins and MRSA isolates on the viability of HT-29 cells. The values were expressed as mean $\pm$ standard error of the mean (SEM) two independent experiments performed in triplicates. The Asterisks (***) indicate significant differences ($p \leq 0.001$) compared to the control.

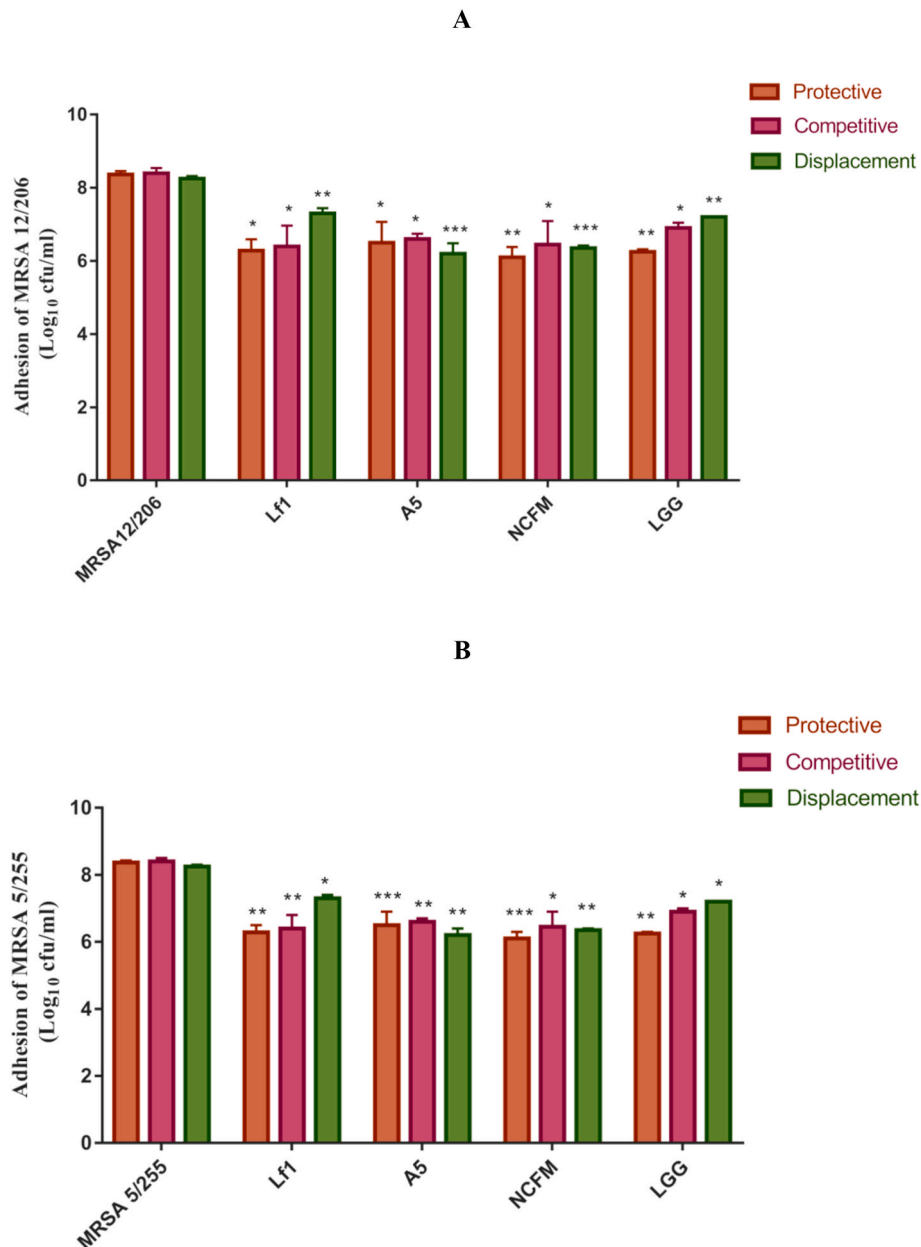


Fig. 3. Anti-mucoadhesion ability of surface proteins against MRSA 12/206 (A) and MRSA 5/255 (B). The values are expressed as mean $\pm$ standard error of the mean (SEM) two independent experiments performed in triplicates. Asterisks (* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$) indicate significant reduction in the MRSA adhesion to the mucin in presence of surface proteins compared to control (12/206 and 5/255).

negatively charged hydrophobic mucin assist the electrostatic interactions [49,50]. Indeed, such interactions assist the self-adhesion of surface proteins with the mucin and thereby mitigate the adhesion of MRSA. In agreement with our findings, the purified surface proteins of probiotic lactobacilli have shown to exclude other mucin adhered enteropathogens [30,39,51].

3.5. Surface proteins interference for adhesion inhibition of MRSA on HT-29 cells

So far, the medical field was using mupirocin and other related antibiotics for the successful decolonization of MRSA [52]. However, the emergence of staphylococcal resistance to such antibiotics calls for unraveling the novel decolonizing agents. In this context, we investigated the anti-adhesion ability of probiotic-derived surface proteins against MRSA isolates through exclusion, competitive, and displacement

assays on HT-29 cells which respectively predict their use as prophylactics, adjuncts, and therapeutics to ameliorate MRSA enterocolitis by hampering the MRSA adhesion to intestinal epithelial cells.

In the protective assay on HT-29 cells, surface proteins of indigenous probiotic strains viz. Lf1 and A5 significantly ($p \leq 0.05$) reduced the adhesion of MRSA 12/206 (8.10 ± 0.10 log₁₀ CFU/mL) to 6.95 ± 0.05 and 6.20 ± 0.05 log₁₀ CFU/mL probably by blocking the adhesion sites (Fig. 4A). Meanwhile, the same strains also rendered anti-adhesion against MRSA 5/255 (8.25 ± 0.05 log₁₀ CFU/mL) by decreasing the adhesion level to 7.03 ± 0.03 (Lf1) and 6.43 ± 0.00 (A5) log₁₀ CFU/mL, respectively. On a similar line, surface proteins of both the reference probiotic strains (NCFM and LGG) also displayed a protective effect against the adhesion of MRSA isolates (12/206 and 5/255) with varying levels of significance as presented in Fig. 4A & B.

Upon testing in competitive adhesion inhibition assay, the surface proteins of Lf1 and A5 competed significantly ($p \leq 0.001$) to the

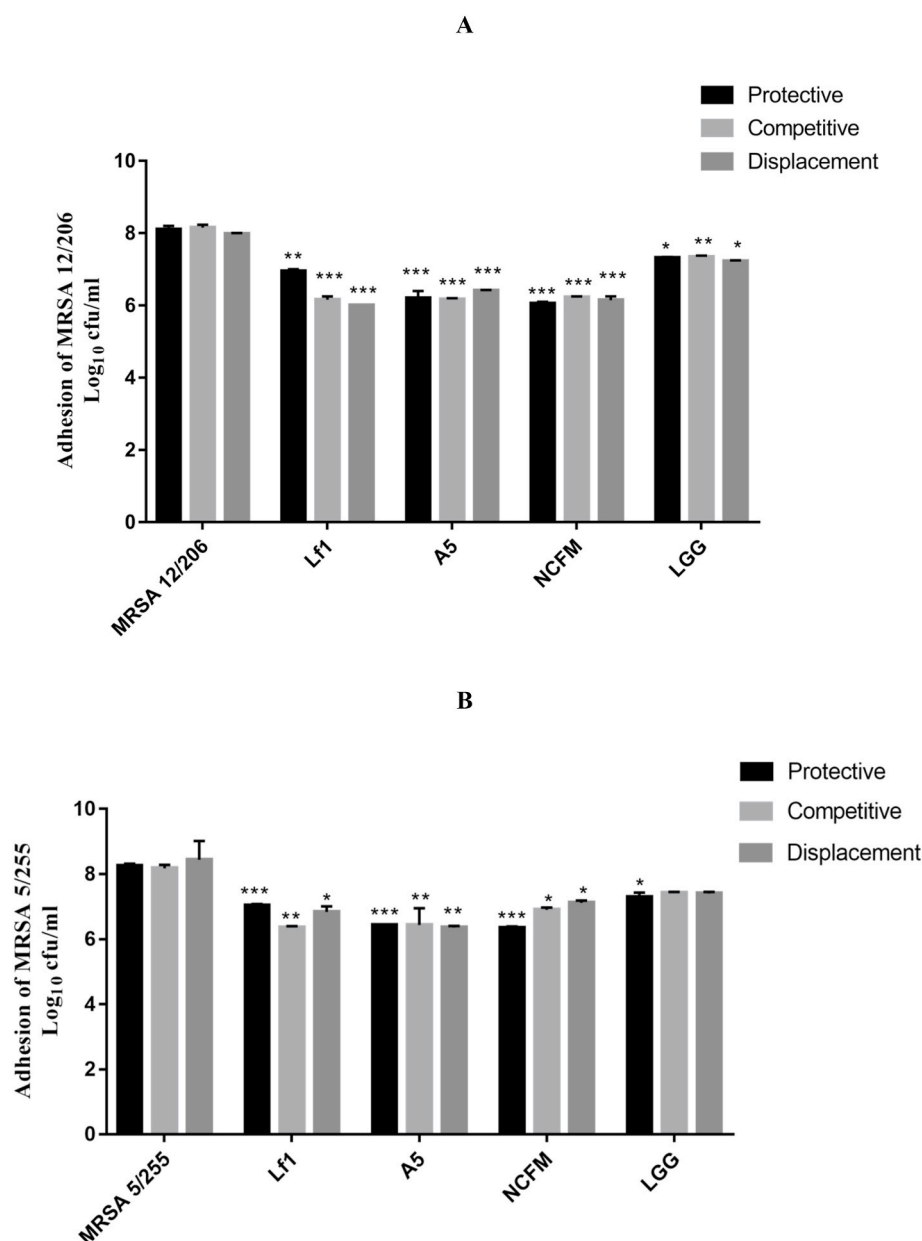


Fig. 4. Anti-adhesion ability of surface proteins against MRSA 12/206 (A) and MRSA 5/255 (B) on HT-29 cell line. The values are expressed as mean $\pm$ standard error of the mean (SEM) two independent experiments performed in triplicates. Asterisks (* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$) indicate significant reduction in the MRSA adhesion to the HT-29 cells in presence of surface proteins compared to control (12/206 and 5/255).

adhesion sites of MRSA 12/206 on HT-29 cells and decreased their adhesion to 6.16 ± 0.09 and 6.17 ± 0.03 log₁₀ CFU/mL, respectively (Fig. 4A). Similarly against 5/255, the significant ($p \leq 0.01$) reduction in adhesion was reported with lower adhesion values of 6.36 ± 0.02 (Lf1) and 6.43 ± 0.36 (A5) log₁₀ CFU/mL (A5) (Fig. 4B). The trivial variations in the antiadhesion ability of surface proteins against two different MRSA isolates possibly indicates the variation in the cell surface proteome (ligand) involved in the interaction with receptor sites on gut enterocytes. Moreover, this also highlights the importance of specific ligand-receptor interaction for effective adhesion and antiadhesion phenomena.

In displacement assay, the surface proteins have displaced the adhered MRSA isolates on the HT-29 cell line in a strain-dependent manner similar to the reference probiotic strains. Post-treatment of surface proteins derived from Lf1 and A5 significantly decreased the adhesion of MRSA 12/206 (7.98 ± 0.02 log₁₀ CFU/mL) to 6.00 ± 0.00 and 6.41 ± 0.02 log₁₀ CFU/mL, respectively (Fig. 5A). Similarly, Lf1 and

A5 could significantly displace the adhered MRSA 5/255 as presented in Fig. 5B. This suggests not only prophylactic but also the therapeutic ability of probiotic-derived surface proteins and their usage would be a noteworthy approach in the treatment of infections caused by gastroenteric pathogens. Although the mechanism of displacement by surface proteins is poorly understood, the surface proteins should either alter the viability of MRSA isolates or should affect the expression of staphylococcal surface adhesion molecules to weaken their subsequent adhesion. In this context, Wang et al. [19] have reported the significant anti-staphylococcal activity of probiotic-derived surface proteins (*L. acidophilus* NCFM and *L. acidophilus* CGMCC 1.1878) (50 μ g/mL) in a combination with nisin as compared to nisin alone. This probably explains that surface proteins might also affect the viability of *S. aureus*.

Earlier, the S-layer protein of *Lactocaseibacillus helveticus* R0052 has shown similar levels of inhibition of *E. coli* O157:H7 to the human epithelial cell line (HEp-2 cell line) [53]. Mucus binding protein extracted from the probiotic *L. plantarum* strain Lp9 successfully

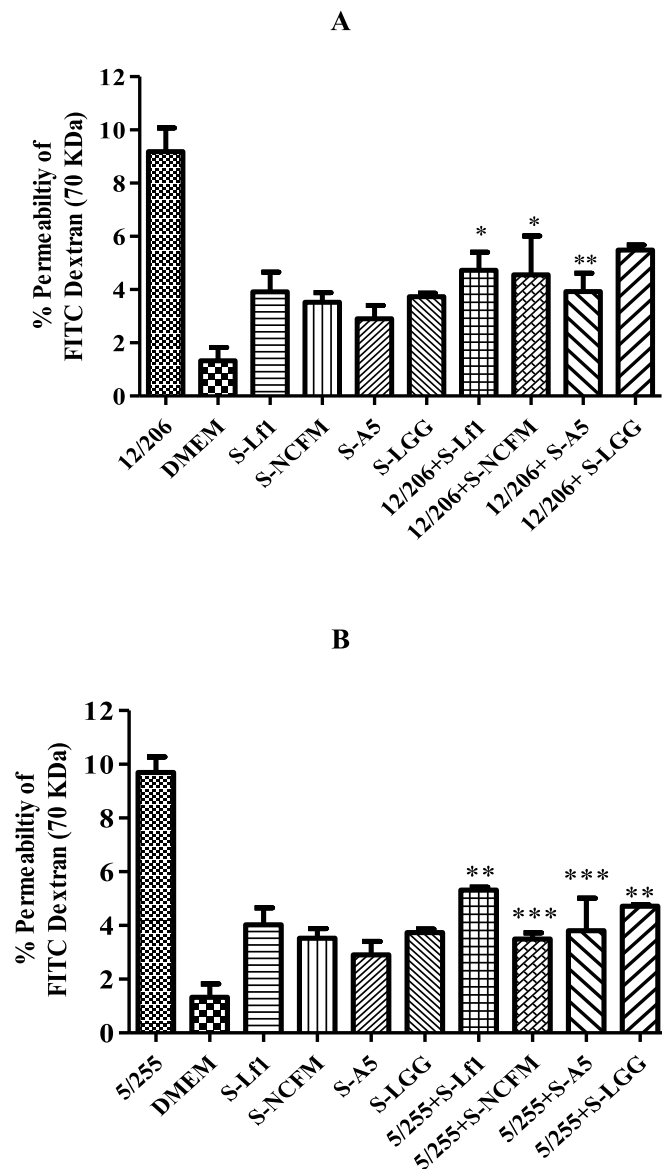


Fig. 5. Effect of surface proteins on epithelial hyperpermeability induced by MRSA 12/206 (A) and 5/255 (B). The values are expressed as mean $\pm$ standard error of the mean (SEM) of two independent experiments performed in triplicates. Asterisks indicate the significant differences: * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$, compared to the control (12/206 and 5/255).

mitigated the adhesion of enterotoxigenic *Escherichia coli* (ETEC) to HT-29 and Caco-2 cell lines by $81 \pm 0.8\%$ and $68 \pm 0.7\%$, respectively [50]. Similar to our findings, Chatterjee et al. [30] have also observed significant anti-adhesion potential of *L. fermentum* derived surface proteins against various enteropathogens. However, the degree of pathogen exclusion by probiotic surface proteins certainly depends on their extent of accessibility to the pattern recognition receptor (PRR) on the epithelial cells.

3.6. Effect of MRSA and surface proteins on FITC-dextran transflux across HT-29 monolayer

The intestinal epithelial architecture comprises enterocytes sealed by tight junction proteins to maintain the normal intestinal permeability for the constant movement of nutrients across the gut barrier. However, such normal permeability gets altered upon direct interaction with pathogens or their toxins leading to impaired intestinal functions with

distinct pathophysiology. Although the mechanism of gut barrier disturbance by gram-negative pathogens is well established [54], the recent findings on staphylococcal mediated damage of intestinal structures via breaching the epithelial integrity highlighted the need for understanding the Gram-positive bacterial mediated ultrastructural changes in the gut epithelium (intestinal disease dynamics) while designing the biotherapeutic agents [9]. Only a handful of studies have shown the *in vitro* protective efficacy of surface proteins of probiotics against gram-negative pathogens induced intestinal barrier dysfunction [55].

HT-29 cells exposed to only MRSA isolates (12/206: $9.18 \pm 0.9\%$ and 5/255: $9.60 \pm 0.5\%$) have shown a tremendous hike in the FITC-dextran diffusion as compared to only DMEM ($1.43 \pm 0.50\%$), thus confirming barrier damage. In agreement with our findings, Temmesfeld-Wollbrück et al. [56] and Kwak et al. [7] have also reported the epithelial hyperpermeability across the Caco-2 cells treated with staphylococcal alpha toxins in time and dosage-dependent fashion. Whereas the cells treated with probiotic surface proteins have shown no significant hike in the dye diffusion as compared to cells treated control (DMEM). On the other hand, the HT-29 cells treated with MRSA isolates in combination with probiotic-derived surface proteins exhibited a significant decrease in the FITC-dextran diffusion as compared to MRSA isolates alone as presented in Fig. 5A and B. In contrast, the surface proteins of LGG showed an insignificant decrease in the dye diffusion induced by MRSA 12/206. This indicates the protective effect of surface proteins of indigenous probiotics against the MRSA isolates induced intestinal hyperpermeability. Previously, Johnson-Henry et al. [53] have reported a significant ($p \leq 0.05$) increase in the TEER value and decrease ($p \leq 0.05$) in the transflux of FITC dextran upon pre-treatment with S-layer proteins of *L. helveticus* R0052 as compared to *E. coli* O157: H7 alone. Surface layer proteins of probiotic *L. plantarum* enhanced the barrier integrity in Caco-2 cells infected with enteropathogenic *E. coli* by significantly $p \leq 0.05$ decreasing the FITC-dextran transflux across the monolayer [57]. In line with our findings, Singh et al. [58] recently reported an approximate 5–6 fold up-regulation in the genes responsible for intestinal epithelial barrier functioning (*hBD-2*, *hBD-3*, *claudin-1*, and *occludin* genes) upon treatment with the cell surface and extracellular proteins (1 $\mu\text{g/mL}$) of *Limosilactobacillus reuteri* strains in *Salmonella typhi* induced Caco-2 model. These evidences suggest the promising effects of probiotic-derived surface proteins in lowering intestinal permeability and thereby improving the intestinal epithelial barrier functioning.

4. Conclusion

In conclusion, the yield and SDS profile of surface proteins derived from probiotic strains under the study were found strain-specific. Surface proteins were found non-cytotoxic to HT-29 cells. The surface proteins of indigenous probiotics significantly hampered the adhesion of MRSA to the HT-29 cells in all three modes of bacterial interference assays. Besides, the co-treatment of MRSA with surface proteins significantly lowered the FITC-dextran transflux across the confluent monolayer of HT-29 cells. Overall, the present study confirmed the prophylactic and therapeutic action of surface proteins of indigenous probiotics against the colonization of MRSA isolates and their subsequent leaky gut infections. However, further preclinical and molecular studies are necessary to validate these findings.

Contributor roles taxonomy (CRediT) author statement

Basavaprabhu Haranahalli Nataraj: Study/Research Execution, Methodology, Software, Formal Analysis, and original draft preparation, Chette Ramesh: Research Execution, Formal analysis, and Validation, Rashmi Hogarehalli Mallappa: Conceptualization, Funding acquisition, Methodology, Supervision, Validation, and Writing - Review & Editing.

Ethical statement approval

The clinical isolates used in the study have been approved by the Institute Ethics Committee of All India Institute of Medical Sciences, New Delhi (Ref No: IEC-484/02.08.2019, RP-54/2019).

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgments

Authors are greatly thankful to the Director, ICAR-National Dairy Research Institute, Karnal for providing the necessary research facilities to carry out the present study. BHN is thankful to the Indian Council of Agricultural Research (ICAR), New Delhi for providing a Junior Research Fellowship (JRF) during his Master's degree program. Authors are thankful to the All India Institute of Medical Sciences (AIIMS), New Delhi for providing clinical isolates of MRSA for research purposes.

References

- G.V. Asokan, T. Ramadhan, E. Ahmed, H. Sanad, WHO global priority pathogens list: a bibliometric analysis of Medline-PubMed for knowledge mobilization to infection prevention and control practices in Bahrain, Oman Med. J. 34 (2019) 184, <https://doi.org/10.5001/omj.2019.37>.
- J.M. Kistler, C.M. Vroome, F.V. Ramsey, A.M. Ilyas, Increasing multidrug antibiotic resistance in MRSA infections of the hand: a 10-year analysis of risk factors, Hand (2019), <https://doi.org/10.1177/1558944719837693>, 1558944719837693.
- D. Wüthrich, A. Cuénod, V. Hinic, M. Morgenstern, N. Khanna, A. Egli, R. Kuehl, Genomic characterization of inpatient evolution of MRSA resistant to daptomycin, vancomycin and ceftaroline, J. Antimicrob. Chemother. 74 (2019) 1452–1454, <https://doi.org/10.1093/jac/dkz003>.
- S. Jayashree, R. Karthikeyan, S. Nithyalakshmi, J. Ranjani, P. Gunasekaran, J. Rajendran, Anti-adhesion property of the potential probiotic strain *Lactobacillus fermentum* 8711 against methicillin-resistant *Staphylococcus aureus* (MRSA), Front. Microbiol. 9 (2018) 411, <https://doi.org/10.3389/fmicb.2018.00411>.
- M. Azimirad, A. Dezfouli, M. Alebouyeh, R.B. Eshfeh, S. Shahrokh, M.R. Zali, Infection with enterotoxigenic *Staphylococcus aureus* as a concern in patients with gastroenteritis, J. Glob. Antimicrob. Resist. 9 (2017) 111–114, <https://doi.org/10.1016/j.jgar.2017.02.009>.
- A.B. Lane, N.K. Copeland, F. Onmus-Leone, J.V. Lawler, Methicillin-Resistant *Staphylococcus aureus* as a probable cause of antibiotic-associated enterocolitis, Case Rep Infect Dis (2018) 1–3, <https://doi.org/10.1155/2018/3106305>.
- Y.K. Kwak, E. Vikström, K.E. Magnusson, B. Vécsey-Semjén, P. Colque-Navarro, R. Möllby, The *Staphylococcus aureus* alpha-toxin perturbs the barrier function in Caco-2 epithelial cell monolayers by altering junctional integrity, Infect. Immun. 80 (2012) 1670–1680, <https://doi.org/10.1128/IAI.00001-12>.
- S. Srisuwan, S.P. Voravuthikunchai, *Rhodomyrtus tomentosa* leaf extract inhibits methicillin-resistant *Staphylococcus aureus* adhesion, invasion, and intracellular survival in human HaCaT keratinocytes, Microb. Drug Resist. 23 (2017) 1002–1012, <https://doi.org/10.1089/mdr.2016.0284>.
- S. Larcombe, J.H. Jiang, M.L. Hutton, H.E. Abud, A.Y. Peleg, D. Lyras, A mouse model of *Staphylococcus aureus* small intestinal infection, J. Med. Microbiol. 69 (2020) 290–297, <https://doi.org/10.1099/jmm.0.001163>.
- M.A. Krezalek, S. Hoyoju, A. Zaborin, E. Okafor, L. Chandrasekar, V. Bindokas, K. Guyton, C.P. Montgomery, R.S. Daum, O. Zaborina, S. Boyle-Vavra, Can methicillin-resistant *Staphylococcus aureus* silently travel from the gut to the wound and cause postoperative infection? Modeling the “Trojan Horse Hypothesis”, Ann. Surg. 267 (2018) 749–758, <https://doi.org/10.1097/SLA.0000000000002173>.
- D.A. Hamza, R.M. Abd-El Salam, S.M. Nader, M. Elhariri, R. Elhelw, H.S. El-Mahallawy, Pathways of methicillin-resistant *Staphylococcus aureus* in animal model: New insights regarding public health, Infect. Drug Resist. 13 (2020) 1593–1600, <https://doi.org/10.2147/IDR.S252332>.
- D.F. Gilpin, S. Small, S. Bakshi, M.P. Kearney, C. Cardwell, M.M. Tunney, Efficacy of a standard methicillin-resistant *Staphylococcus aureus* decolonisation protocol in routine clinical practice, J. Hosp. Infect. 75 (2010) 93–98, <https://doi.org/10.1016/j.jhin.2009.11.025>.
- D.S. Acton, M.T. Plat-Sinnige, W. van Wamel, N. de Groot, A. van Belkum, Intestinal carriage of *Staphylococcus aureus*: how does its frequency compare with that of nasal carriage and what is its clinical impact? Eur. J. Clin. Microbiol. Infect. Dis. 28 (2009) 115, <https://doi.org/10.1007/s10096-008-0602-7>.
- C. Ghosh, P. Sarkar, R. Issa, J. Haldar, Alternatives to conventional antibiotics in the era of antimicrobial resistance, Trends Microbiol. 27 (2019) 323–338, <https://doi.org/10.1016/j.tim.2018.12.010>.
- A. Henein, What are the limitations on the wider therapeutic use of phage? Bacteriophage 3 (2013), e24872 <https://doi.org/10.4161/bact.24872>.
- J.E. Aguilar-Toalá, R. García-Varela, H.S. García, V. Mata-Haro, A.F. González-Córdova, B. Vallejo-Cordoba, A. Hernández-Mendoza, Postbiotics: an evolving term within the functional foods field, Trends Food Sci. Technol. 75 (2018) 105–114, <https://doi.org/10.1016/j.tifs.2018.03.009>.
- P.F. Cuevas-González, A.M. Liceaga, J.E. Aguilar-Toalá, Postbiotics and Paraprobiotics: from concepts to applications, Food Res. Int. 136 (2020), 109502, <https://doi.org/10.1016/j.foodres.2020.109502>.
- F.L. do Carmo, H. Rabah, R.D. De Oliveira Carvalho, F. Gaucher, B.F. Cordeiro, S. H. da Silva, Y. Le Loir, V. Azevedo, G. Jan, Extractable bacterial surface proteins in probiotic–host interaction, Front. Microbiol. 9 (2018) 645, <https://doi.org/10.3389/fmicb.2018.01807>.
- H. Wang, Y. Niu, J. Pan, Q. Zhang, R. Lu, Antibacterial effects of *Lactobacillus acidophilus* surface-layer protein in combination with nisin against *Staphylococcus aureus*, LWT - Food Sci. Technol. (Lebensmittel-Wissenschaft - Technol.) 124 (2020), 109208, <https://doi.org/10.1016/j.lwt.2020.109208>.
- T.P. Singh, G. Kaur, S. Kapila, R.K. Malik, Antagonistic activity of *Lactobacillus reuteri* strains on the adhesion characteristics of selected pathogens, Front. Microbiol. 8 (2017) 486, <https://doi.org/10.3389/fmicb.2017.00486>.
- G. Wang, M. Zhang, J. Zhao, Y. Xia, P.F.H. Lai, L. Ai, A surface protein from *Lactobacillus plantarum* increases the adhesion of lactobacillus strains to human epithelial cells, Front. Microbiol. 9 (2018) 2858, <https://doi.org/10.3389/fmicb.2018.02858>.
- F.L. do Carmo, H. Rabah, S. Huang, F. Gaucher, M. Deplanche, S. Dutertre, J. Jarden, Y. Le Loir, V. Azevedo, G. Jan, *Propionibacterium freudenreichii* surface protein SlpB is involved in adhesion to intestinal HT-29 cells, Front. Microbiol. 8 (2017) 1033, <https://doi.org/10.3389/fmicb.2017.01033>.
- V. Dubey, A.K. Mishra, A.R. Ghosh, Cell adherence efficacy of probiotic *Pediococcus pentosaceus* GS4 (MTCC 12683) and demonstrable role of its surface layer protein (Slp), J. Proteom. 226 (2020), 103894, <https://doi.org/10.1016/j.jpro.2020.103894>.
- C. Klotz, Y.J. Goh, S. O'Flaherty, B. Johnson, R. Barrangou, Deletion of S-layer associated Ig-like domain protein disrupts the *Lactobacillus acidophilus* cell surface, Front. Microbiol. 11 (2020) 345, <https://doi.org/10.3389/fmicb.2020.00345>.
- B.H. Nataraj, C. Ramesh, R.H. Mallappa, Characterization of antibiotic resistance and virulence traits present in clinical methicillin-resistant *Staphylococcus aureus* isolates, Curr. Microbiol. 1–14 (2021), <https://doi.org/10.1007/s00284-021-02477-x>.
- M.S. Turner, P. Timms, L.M. Hafner, P.M. Giffard, Identification and characterization of a basic cell surface-associated protein from *Lactobacillus fermentum* BR11, J. Bacteriol. 179 (1997) 3310–3316, <https://doi.org/10.1128/jb.179.10.3310-3316.1997>.
- O.H. Lowry, N.J. Rosebrough, A.L. Farr, R.J. Randall, Protein measurement with the Folin phenol reagent, J. Biol. Chem. 193 (1951) 265–275.
- U.K. Laemmli, Cleavage of structural proteins during the assembly of the head of bacteriophage T4, Nature 227 (1970) 680–685.
- M. Thirabunyanon, P. Boonprasom, P. Niamsup, Probiotic potential of lactic acid bacteria isolated from fermented dairy milks on antiproliferation of colon cancer cells, Biotechnol. Lett. 31 (2009) 571–576, <https://doi.org/10.1007/s10529-008-9902-3>.
- M. Chatterjee, A.C. Pushkaran, A.K. Vasudevan, K.K.N. Menon, R. Biswas, C. G. Mohan, Understanding the adhesion mechanism of a mucin binding domain from *Lactobacillus fermentum* and its role in enteropathogen exclusion, Int. J. Biol. Macromol. 110 (2018) 598–607, <https://doi.org/10.1016/j.ijbiomac.2017.10.107>.
- P. Palumbo, F. Lombardi, M.G. Cifone, B. Cinque, The epithelial barrier model shows that the properties of VSL# 3 depend from where it is manufactured, Endocr. Metab. Immune Disord. - Drug Targets 19 (2019) 199–206, <https://doi.org/10.2174/1871530318666181022164505>.
- I.A. van Dijk, M.L. Ferrando, A.E. van der Wijk, R.A. Hoebe, K. Nazmi, W.J. de Jonge, P.M. Krawczyk, J.G. Bolscher, E.C. Veerman, J. Stap, Human salivary peptide histatin-1 stimulates epithelial and endothelial cell adhesion and barrier function, FASEB. J. 31 (2017) 3922–3933, <https://doi.org/10.1096/fj.201700180R>.
- M.I. Bhat, K. Sowmya, S. Kapila, R. Kapila, Potential probiotic *Lactobacillus rhamnosus* (MTCC-5897) inhibits *Escherichia coli* impaired intestinal barrier function by modulating the host tight junction gene response, Probiotics Antimicrob. Proteins 1 (2019) 12, <https://doi.org/10.1007/s12602-019-09608-8>.
- C. Ghosh, P. Sarkar, R. Issa, J. Haldar, Alternatives to conventional antibiotics in the era of antimicrobial resistance, Trends Microbiol. 27 (2019) 323–338, <https://doi.org/10.1016/j.tim.2018.12.010>.
- T.P. Singh, R.K. Malik, G. Kaur, Cell surface proteins play an important role in probiotic activities of *Lactobacillus reuteri*, Nutrire 41 (2016) 5, <https://doi.org/10.1186/s41110-016-0007-9>.
- Y. Zhang, X. Xiang, Q. Lu, L. Zhang, F. Ma, L. Wang, Adhesions of extracellular surface-layer associated proteins in *Lactobacillus* M5-L and Q8-L, J. Dairy Sci. 99 (2016) 1011–1018, <https://doi.org/10.3168/jds.2015-10020>.
- B. Sánchez, P. Bressollier, S. Chaignepain, J.M. Schmitter, M.C. Urdaci, Identification of surface-associated proteins in the probiotic bacterium *Lactobacillus rhamnosus* GG, Int. Dairy J. 19 (2009) 85–88, <https://doi.org/10.1016/j.idairyj.2008.09.005>.
- M. Rojas, F. Ascencio, P.L. Conway, Purification and characterization of a surface protein from *Lactobacillus fermentum* 104R that binds to porcine small intestinal mucus and gastric mucin, Appl. Environ. Microbiol. 68 (2002) 2330–2336, <https://doi.org/10.1128/AEM.68.5.2330-2336.2002>.
- A.K. Yadav, A. Tyagi, J.K. Kaushik, A.C. Saklani, S. Grover, V.K. Batish, Role of surface layer collagen binding protein from indigenous *Lactobacillus plantarum* 91 in adhesion and its anti-adhesion potential against gut pathogen, Microbiol. Res. 168 (2013) 639–645, <https://doi.org/10.1016/j.micres.2013.05.003>.
- M. Ishida, F. Namai, S. Shigemori, S. Kajikawa, M. Tsukagoshi, T. Sato, T. Ogita, T. Shimamoto, Ribosome-engineered *Lactocaseibacillus rhamnosus* strain GG exhibits

- cell surface glyceraldehyde-3-phosphate dehydrogenase accumulation and enhanced adhesion to human colonic mucin, *Appl. Environ. Microbiol.* 86 (2020), <https://doi.org/10.1128/AEM.01448-20> e01448-20.
- [41] J. Frece, B. Kos, I.K. Svetec, Z. Zgaga, V. Mrša, J. Šušković, Importance of S-layer proteins in probiotic activity of *Lactobacillus acidophilus* M92, *J. Appl. Microbiol.* 98 (2005) 285–292, <https://doi.org/10.1111/j.1365-2672.2004.02473.x>.
- [42] S. Bisht, K.S. Singh, R. Choudhary, S. Kumar, S. Grover, A.K. Mohanty, V. Pande, J. K. Kaushik, Expression of fibronectin-binding protein of *L. acidophilus* NCFM and *in vitro* refolding to adhesion capable native-like protein from inclusion bodies, *Protein Expr. Purif.* 145 (2018) 7–13, <https://doi.org/10.1016/j.pep.2017.11.007>.
- [43] M.E. Macías-Rodríguez, M. Zagorec, F. Ascencio, R. Vázquez-Juárez, M. Rojas, *Lactobacillus fermentum* BCS87 expresses mucus-and mucin-binding proteins on the cell surface, *J. Appl. Microbiol.* 107 (2009) 1866–1874, <https://doi.org/10.1111/j.1365-2672.2009.04368.x>.
- [44] C.J. Jeffery, Intracellular/surface moonlighting proteins that aid in the attachment of gut microbiota to the host, *AIMS Microbiol.* 5 (2019) 77.
- [45] L. Muscarello, B. De Siena, R. Marasco, *Lactobacillus* cell surface proteins involved in interaction with mucus and extracellular matrix components, *Curr. Microbiol.* (2020) 1–11, <https://doi.org/10.1007/s00284-020-02243-5>.
- [46] M. Ishiyama, H. Tominaga, M. Shiga, K. Sasamoto, Y. Ohkura, K. Ueno, A combined assay of cell viability and *in vitro* cytotoxicity with a highly water-soluble tetrazolium salt, neutral red and crystal violet, *Biol. Pharm. Bull.* 19 (1996) 1518–1520.
- [47] B.B. Zhao, J. Meng, Q.X. Zhang, T.T. Kang, R.R. Lu, Protective effect of surface layer proteins isolated from four *Lactobacillus* strains on hydrogen-peroxide-induced HT-29 cells oxidative stress, *Int. J. Biol. Macromol.* 102 (2017) 76–83, <https://doi.org/10.1016/j.ijbiomac.2017.03.160>.
- [48] S. Nozari, Y. Faridvand, A. Etesami, M. Ahmad Khan Beiki, S.A. Miresmaeili Mazraekhondi, J. Abdolalizadeh, Potential anticancer effects of cell wall protein fractions from *Lactobacillus paracasei* on human intestinal Caco-2 cell line, *Lett. Appl. Microbiol.* 69 (2019) 148–154, <https://doi.org/10.1111/lam.13198>.
- [49] M.L. Van Tassel, M.J. Miller, *Lactobacillus* adhesion to mucus, *Nutrients* 3 (2011) 613–636, <https://doi.org/10.3390/nu3050613>.
- [50] K.S. Singh, S. Kumar, A.K. Mohanty, S. Grover, J.K. Kaushik, Mechanistic insights into the host-microbe interaction and pathogen exclusion mediated by the Mucus-binding protein of *Lactobacillus plantarum*, *Sci. Rep.* 8 (2018) 1–10, <https://doi.org/10.1038/s41598-018-32417-y>.
- [51] L. Du, F. Liu, X. Ju, G. Huo, Adhesion capability of first two domains at N terminus of NP_785232 protein and their interaction with a UV-absorbing component from human mucus, *Lett. Appl. Microbiol.* 51 (2010) 400–405, <https://doi.org/10.1111/j.1472-765X.2010.02911.x>.
- [52] T.T. Poovelikunnel, P.E. Budri, A.C. Shore, D.C. Coleman, H. Humphreys, D. Fitzgerald-Hughes, Molecular characterization of nasal methicillin-resistant *Staphylococcus aureus* isolates showing increasing prevalence of mupirocin resistance and associated multidrug resistance following attempted decolonization, *Antimicrob. Agents Chemother.* 62 (2018), <https://doi.org/10.1128/AAC.00819-18> e00819-18.
- [53] K.C. Johnson-Henry, K.E. Hagen, M. Gordonpour, T.A. Tompkins, P.M. Sherman, Surface-layer protein extracts from *Lactobacillus helveticus* inhibit enterohaemorrhagic *Escherichia coli* O157: H7 adhesion to epithelial cells, *Cell Microbiol.* 9 (2007) 356–367, <https://doi.org/10.1111/j.1462-5822.2006.00791.x>.
- [54] M.I. Bhat, K. Sowmya, S. Kapila, R. Kapila, *Escherichia coli* K12: an evolving opportunistic commensal gut microbe distorts barrier integrity in human intestinal cells, *Microb. Pathog.* 133 (2019) 103545, <https://doi.org/10.1016/j.micpath.2019.103545>.
- [55] H. Wang, Q. Zhang, Y. Niu, X. Zhang, R. Lu, Surface-layer protein from *Lactobacillus acidophilus* NCFM attenuates tumor necrosis factor- α -induced intestinal barrier dysfunction and inflammation, *Int. J. Biol. Macromol.* 136 (2019) 27–34, <https://doi.org/10.1016/j.ijbiomac.2019.06.041>.
- [56] B. Temmesfeld-Wollbrück, B. Brell, C. zu Dohna, M. Dorenberg, A.C. Hocke, H. Martens, J. Klar, N. Suttrop, S. Hippenstiel, Adrenomedullin reduces intestinal epithelial permeability *in vivo* and *in vitro*, *Am. J. Physiol. Gastrointest. Liver Physiol.* 297 (2009) G43–G51, <https://doi.org/10.1152/ajpgi.90532.2008>.
- [57] Z. Liu, T. Shen, P. Zhang, Y. Ma, H. Qin, *Lactobacillus plantarum* surface layer adhesive protein protects intestinal epithelial cells against tight junction injury induced by enteropathogenic *Escherichia coli*, *Mol. Biol. Rep.* 38 (2011) 3471–3480, <https://doi.org/10.1007/s11033-010-0457-8>.
- [58] T.P. Singh, N. Tehri, G. Kaur, R.K. Malik, Cell surface and extracellular proteins of potentially probiotic *Lactobacillus reuteri* as an effective mediator to regulate intestinal epithelial barrier function, *Arch. Microbiol.* 1–10 (2021), <https://doi.org/10.1007/s00203-021-02318-2>.